Ideal Facial Ratios and Angles for Men and Women: What the Literature Actually Supports

Only about half of these 16 metrics have real peer-reviewed reference values: nasofrontal angle, submental-cervical angle, facial convexity, total facial convexity, mandibular plane angle, gonial angle, canthal tilt, facial thirds and eye spacing. Of those, only canthal tilt, gonial angle, facial thirds, the nasofrontal angle and the mandibular plane angle show a consistent difference between men and women. "Jaw frontal angle", "ramus to mandible ratio", "total FWHR", "cheekbone height %" and "eye separation ratio" as usually quoted come mostly from looksmaxxing software such as FaceIQ Labs, not from clinical norms.

TL;DR

- **Best-supported numbers:**
  - Nasofrontal angle: 115–130° (Powell & Humphreys)
  - Submental-cervical angle: 105–120° (Ellenbogen & Karlin, 1980)
  - Facial convexity (G–Sn–Pg): about 165–175°, with soft-tissue norms near 168–169° for both sexes (Arnett 1999; Fernández-Riveiro 2003)
  - Total facial convexity (G–Prn–Pg): about 138.6° in men vs 142° in women (Fernández-Riveiro 2003)
  - Mandibular plane angle: SN–GoGn 32° (Steiner) or FMA 25° (Tweed)
  - Canthal tilt: about 4° in average Caucasians, and 6.5° (men) vs 8.5° (women) in attractive celebrities (Liu et al. 2023)
- **Where men and women really differ:**
  - Women have about 3–5° wider gonial angles.
  - Women have steeper canthal tilt and a more obtuse nasofrontal angle.
  - Men have a larger lower facial third.
  - FWHR differences between the sexes are small and inconsistent once body size is controlled.
- **Treat the other metrics as software conventions:** jaw frontal angle, cheekbone height %, ramus/mandible ratio, total FWHR and eye separation ratio. The "golden ratio" (1.618) works only for selected ratios, e.g., facial height ÷ jaw width ≈ 1.61 in attractive women. In Pallett, Link & Lee's 2010 *Vision Research* experiments, attractiveness peaked at the ratios of an average face (eye–mouth distance 36% of face length, interocular distance 46% of face width), not at phi.

Summary Table

Measurement	Male range	Female range	Notes / source
Golden ratio (phi = 1.618)	Facial height ÷ bigonial width ≈ 1.566 ± 0.085	≈ 1.613 ± 0.063	Liu, Salinas & Sharaf 2023 (*Aesthet Surg J Open Forum*), 42 Caucasian celebrities. Pallett, Link & Lee 2010 (*Vision Research*): the "new golden ratios" of 36% (eye–mouth ÷ face length) and 46% (interpupillary distance ÷ face width) match the average face, not phi. nih +2
Gonial angle (Ar–Go–Me)	Mean about 121–126°	About 125–129° (≈3–5° higher)	Jensen & Palling 1954 ("The gonial angle: a survey," *Am J Orthod*): range 100–148°, and females 3–5° higher in all racial groups. CBCT data: 125.7° (M) vs 128.7° (F). nih CT data: 121.6° vs 125.2°. sciencedirect No validated "attractive" range exists.
Facial thirds	≈31.0 / 30.5 / 38.5%	≈29.5 / 32.4 / 38.2%	Farkas norms (trichion–nasion / nasion–subnasale / subnasale–gnathion), as cited by Villavisanis et al. 2022. upenn The classical canon is equal thirds (33/33/33). qoves Attractive women have more even thirds (Liu 2023). researchgate FaceIQ (non-academic): lower third 33.9–37.0% (M) vs 30.0–33.1% (F). faceiqlabs faceiqlabs
Eye separation ratio (IPD ÷ face width, assumed)	≈46%	≈46%	Pallett et al. 2010 (female faces only): optimum ≈46%. eurekalert perfectcorp FaceIQ (non-academic): 45.7–46.8% (M), 46.4–47.5% (F). faceiqlabs faceiqlabs
Total facial convexity (G–Prn–Pg, nose included)	138.6 ± 6.8°	142.0 ± 6.1°	Fernández-Riveiro et al. 2003 (*Eur J Orthod*), Spanish young adults. apospublications Higher in southern Chinese youth (≈147–149°). nih Bangladeshi adults ≈127° when measured from nasion (N–Prn–Pg). apospublications

Measurement	Male range	Female range	Notes / source
Facial convexity (G–Sn–Pg, nose excluded)	≈165–175°; Arnett norm 169.4 ± 3.2°	≈165–175°; Arnett norm 169.3 ± 3.4°	Arnett et al. 1999 (*AJODO*) soft-tissue cephalometric analysis (STCA). sagepub Bergman longitudinal data: 169° (M) and 168° (F) at age 18. nih North Indians ≈165°. Essentially no sex difference.
Cheekbone position	Malar peak ≈2 mm ahead of the cornea; above/behind Hinderer's lines; ⅓ down from the lateral canthus (Wilkinson)	Same landmarks; women's cheekbones relatively more prominent	Hinderer 1975 (*Plast Reconstr Surg*). intechopen Nechala, Mahoney & Farkas 2000 found Powell's point 28 mm too lateral. sagepub Bizygomatic/bigonial width 1.128 (M) vs 1.174 (F) (Liu 2023). FaceIQ "cheekbone height" (non-academic): 83–100% (M), 84–101% (F). faceiqlabs faceiqlabs
Jaw frontal angle (non-standard)	86.5–92.5°	87.5–93.5°	FaceIQ Labs only (non-academic). faceiqlabs +2 Not found in the clinical literature.
Mandibular plane angle	SN–GoGn 32° (normal 28–36°); FMA 25° (normal 20–30°)	Same norms; women typically a few degrees steeper	Steiner, Tweed and Downs (MP/SN 27–37°). slideshare +4 Males showed lower SN–GoGn/FMA than females (Philadelphia African-American norms, 2025). springer
Canthal tilt	Average ≈4°; attractive celebrities 6.51 ± 2.67°	Average ≈4°; attractive celebrities 8.50 ± 2.10°	Liu et al. 2023, citing Rhee et al. 2012 (4.12° in the average Caucasian composite). nih scienceinsights Rhee: 4.84–5.86° in "beautiful" faces. researchgate
Ramus to mandible ratio (non-standard)	≈0.62 (Jarabak ≈44 mm ÷ 71 mm)	Similar ratio; absolute lengths smaller	Björk–Jarabak cephalometric norms. Ramus height and body length are larger in men (Saudi/Kurdish samples). edu FaceIQ (non-academic): 0.6–0.8 (internal 0.62–0.75). faceiqlabs

Measurement	Male range	Female range	Notes / source
FWHR (bizygomatic width ÷ brow–upper lip)	Means ≈1.8–1.95	Means ≈1.7–1.9	Carré & McCormick 2008: ≈1.86 (M) vs ≈1.80 (F), reported secondarily. realsmile Kramer 2015 athletes: hockey 1.91 vs 1.90. doi Lefevre et al. 2012 (*Evol Hum Behav*) found no significant sex difference in three of four samples, and Kramer 2017 found none in White faces (d = −0.05). No "ideal" exists.
Total FWHR (non-standard)	≈1.30–1.40 (FaceIQ)	Not established	FaceIQ only (non-academic), probably bizygomatic width ÷ brow-to-chin height. faceiqlabs Comparable academic ratio: facial height/bizygomatic width 1.390 (M) vs 1.375 (F), no significant difference (Liu 2023). oup
Submental-cervical angle	105–120° (youthful ideal); population mean ≈126°	105–120°; population mean ≈121°	Ellenbogen & Karlin 1980 (*Plast Reconstr Surg*). ovid Later analysis: 90–120°. oup The Sommerville means come from a secondary source (Qoves). qoves Different from Powell's mentocervical angle (80–95°). brainscape
Nasofrontal angle (G–N–nasal tip/dorsum)	115–130°, toward the acute end; Turkish mean 123.9 ± 13.2°	115–130°, toward the obtuse end; Turkish mean 133.2 ± 8.9°	Powell & Humphreys 1984. rhinoplastyarchive Uzun et al. 2014 (as cited by Qoves). qoves Southern Chinese 137.9 ± 9.4°. liebertpub Wider in East Asians. researchgate
Eye spacing (intercanthal distance, en–en; IPD)	ICD ≈30–35 mm; IPD 64.0 ± 3.4 mm	ICD ≈30–35 mm; IPD 61.7 ± 3.6 mm	Caucasian ICD norm 30–35 mm (Farkas tradition). cibtech Black adults ICD 33.9 ± 3.0 mm. sciencedirect 2012 US Army anthropometric survey (IPD). Wikipedia Neoclassical canon: ICD ≈ eye width ≈ nose width. researchgate

Key Findings

1. The strongest, best-replicated "ideals" come from surgery and orthodontics.

- **Powell & Humphreys aesthetic triangle (1984):**
  - nasofrontal 115–130°
  - nasofacial 30–40°
  - nasomental 120–132°
  - mentocervical 80–95°
  - These remain the standard Caucasian references. One rhinoplasty text adds that, within 115–130°, a more obtuse nasofrontal angle suits women and a more acute one suits men. rhinoplastyarchive
- **Ellenbogen & Karlin (1980):** a cervicomental angle of 105–120° is still the most cited marker of a youthful neck. sagepub Angles above about 120° read as a heavy neck or "double chin." plasticsurgerykey qoves
- **Profile convexity:** Arnett's soft-tissue facial angle and the Fernández-Riveiro photogrammetric norms show that the profile without the nose (G–Sn–Pg) is nearly identical in men and women (≈168–169°). The angle including the nose (G–Prn–Pg) is about 3° more open in women (142° vs 138.6°), apospublications reflecting smaller noses and straighter profiles in women.

2. Sexual dimorphism is real but modest, and concentrated in a few metrics.

- **Gonial angle:** Jensen & Palling's 1954 survey (*Am J Orthod*) reported that "in all racial groups the mean angle in females is 3 degrees to 5 degrees higher than in males." Recent CBCT/CT studies agree, at about 3–3.5°. sciencedirect nih Individual variation (100–148°), however, swamps the sex difference.
- **Canthal tilt:** the attractive-celebrity data show women about 2° steeper than men (8.5° vs 6.5°), and both well above the ≈4° population average. This suggests a modest positive tilt is attractive in both sexes, and more so in women. nih scienceinsights
- **Lower face:** Arnett's model data show male faces are longer overall (soft-tissue nasion–menton about 13 mm longer), including a lower third about 10 mm longer. uspto Farkas-style percentages give men a slightly smaller middle third and slightly larger upper and lower thirds. upenn

3. FWHR is a behavioral-perception metric, not an attractiveness standard.

- **Origin:** Weston, Friday & Liò (2007) proposed FWHR as sexually dimorphic based on 121 southern African skulls. sciencedirect Carré & McCormick (2008) linked it to aggression in men. softmaxx researchgate
- **Failed replication:** Lefevre et al. (2012, *Evol Hum Behav*) found no significant sex difference in fWHR in three of four large adult samples, even though other facial metrics showed large sex differences in all four. Kramer's 2017 meta-analysis (*Evol Hum Behav*)

found a small male-larger effect in skulls but no evidence of sex differences in White faces (N = 2,037, k = 9, mean weighted d = −0.05).

- **Bizygomatic width:** Caton & Dixson (2022, *Biology Letters*) studied 6,068 US Army personnel (ANSUR II) and found bizygomatic width itself is strongly dimorphic (d = 1.39), staying large (d = 1.07) after controlling for allometry.
- **What this means:** a "high FWHR" (≈1.9–2.0+) signals perceived dominance more than attractiveness. qoves Any "ideal FWHR" figure is a forum convention, not a research finding. symmiapp qoves

4. The "golden ratio" holds only where someone picked the right ratio.

- Liu et al. found facial height ÷ bigonial width = 1.613 in attractive women, close to phi. nih
- Pallett et al. ran controlled experiments and found the most attractive configuration matches the *average* face: eye–mouth distance ≈36% of face length, interpupillary distance ≈46% of face width. perfectcorp They reported that these optima were significantly different from the classic golden-ratio value of .38 (Pallett, Link & Lee 2010, *Vision Research*).
- **Conclusion:** use averageness-based ratios, not phi, as the evidence-backed target.

Details by Measurement (definitions and landmarks)

- **Golden ratio:** any facial ratio compared with 1.618. There is no single standard definition.
  - Most defensible operational versions: facial height (trichion or nasion to menton) ÷ bigonial width (Liu 2023), or Pallett's two averageness ratios.
  - Treat phi-based "beauty masks" (e.g., Marquardt) as non-academic.
- **Gonial angle:** the angle at gonion between the ramus tangent (articulare–gonion) and the mandibular lower border (gonion–menton), measured on cephalograms, panoramic radiographs or CBCT.
  - The common clinical norm is about 130° ± 7° (Björk–Jarabak), and adult means of 120–130° are typical.
  - Lower values give a squarer, more "masculine" jaw. The preference for ~110–120° on looksmaxxing forums has no validated basis.
- **Facial thirds:** trichion–glabella (or nasion), glabella–subnasale, and subnasale–menton (or gnathion).
  - The classical ideal is equal thirds. springer
  - Measured norms show the lower third is usually the largest in both sexes. researchgate

    - Fernández-Riveiro's absolute values are similar for both thirds within each sex: G–Sn 72.1 mm (M) vs 68.7 mm (F), and Sn–Me 71.4 mm (M) vs 65.4 mm (F). ovid
- **Eye separation ratio (most likely interpretation):** interpupillary distance ÷ bizygomatic face width. perfectcorp This matches Pallett's "interocular distance ÷ face width" and the probable FaceIQ definition, but FaceIQ does not publish its formula, so this is an assumption.
- **Eye spacing:** absolute intercanthal distance (en–en) and/or interpupillary distance (pupil to pupil).
    - The neoclassical canon says intercanthal width should equal one eye-fissure width and the alar (nose) width. researchgate
    - Attractive Korean faces have relatively wide-set eyes (Rhee 2012). Korean pageant contestants nonetheless had an absolute ICD of about 34.3 ± 1.86 mm. springer springer
- **Total facial convexity:** soft-tissue glabella → pronasale (nose tip) → soft-tissue pogonion, so the nose is included. researchgate
    - Some authors measure from nasion (N–Prn–Pg). That gives about 127–140°, so always check the vertex landmarks. apospublications ovid
- **Facial convexity:** soft-tissue glabella → subnasale → soft-tissue pogonion, with the nose excluded. speareducation
    - The angle is usually reported as the obtuse interior angle (≈165–175°). speareducation
    - Arnett/Legan-style analyses sometimes report its supplement (≈12–14° of convexity), which is why Turkish norms appear as "14°".
- **Cheekbone position:** there is no single angle or ratio.
    - Clinical methods: Hinderer's intersection (lateral canthus–oral commissure line × tragus–alar base line, with the prominence posterosuperior to the crossing), Wilkinson's ⅓-down-from-canthus point, and the sagittal rule that the malar prominence sits about 2 mm anterior to the cornea (a "negative vector" if the globe is ahead). intechopen +2
    - Frontally, a bizygomatic width larger than bigonial width tapers the face, more so in women.
    - FaceIQ's "cheekbone height %" appears to follow a forum convention (cheekbone line height relative to the pupil line) and is unvalidated. looksmax
- **Jaw frontal angle:** the definition is not published. It appears to be a frontal-view angle at the jaw contour, near gonion, between the lower jaw border and the chin/face line.
    - A 2023 survey of attractive white female jaws reported a frontal gonial angle of about 142°, clearly measured differently. sciencedirect

    - Treat FaceIQ's ≈87–93° figures as internal software parameters only. faceiqlabs faceiqlabs
- **Mandibular plane angle:**
    - Steiner: S–N to Go–Gn, 32° (normodivergent ≈28–36°) slideshare Wikipedia
    - Tweed: FMA, Frankfort horizontal (porion–orbitale) to Go–Me, 25° (clinical 20–30°) bceph uspto
    - Downs: MP to FH, ≈26° (alternate MP/SN formulation 27–37°) nih
    - Higher angles mean a long-face, hyperdivergent pattern. bceph researchgate
- **Canthal tilt (palpebral axis):** the angle between the line from medial canthus (endocanthion) to lateral canthus (exocanthion) and true horizontal. oup realsmile Positive means the lateral canthus is higher.
    - A popular secondary source attributes about +3° for men to Volpe & Ramirez (2005), pinkmirror but it has not been verified here.
- **Ramus to mandible ratio (most likely interpretation):** ramus height (Ar–Go or condylion–gonion) ÷ mandibular body length (Go–Me or Go–Gn).
    - Björk–Jarabak norms of ≈44 mm ramus height and ≈71 ± 5 mm body length imply about 0.62. researchgate
    - A clinically related index is Jarabak's posterior/anterior face height ratio (S–Go ÷ N–Me) of 62–65%. Below 62% indicates vertical growth, above 65% horizontal growth. scribd bceph
- **FWHR:** bizygomatic width (zygion–zygion) ÷ upper-face height, defined as upper lip (labrale superius or prosthion) to brow or nasion. sagepub qoves
    - Skull studies use nasion–prosthion; photo studies use brow–lip. plos
    - The two conventions do not give interchangeable numbers.
- **Total FWHR:** most likely bizygomatic width ÷ full face height (brow, glabella or nasion to menton), giving values of about 1.3–1.4. Some apps instead invert it (height ÷ width).
- **Submental-cervical angle:** the cervicomental angle between a line tangent to the submental (under-chin) plane through menton and a line tangent to the anterior neck, with the vertex near the hyoid. researchgate plasticsurgerykey Distinguish it from:
    - Powell's mentocervical angle: glabella–pogonion line vs cervical point–menton line, 80–95° brainscape
    - The photogrammetric "cervicomental" angle G–Pg/C–Me nih
- **Nasofrontal angle:** at soft-tissue nasion, between the nasion–glabella line and the nasion–nasal dorsum (or tip) line. rhinoplastyarchive

Ethnic Variation

- **Noses and the nasofrontal angle:** Chinese and other East Asian faces have wider nasofrontal angles (southern Chinese ≈138° vs the 115–130° Caucasian ideal). researchgate liebertpub East Asian aesthetic preferences also favor a more obtuse nasofrontal angle and a slightly retruded mandible. sciencedirect
- **Arnett's norms:** they do not transfer across populations. researchgate North Indian facial angles (≈165°) were significantly lower than Arnett's 169° Caucasian norm. sagepub
- **Gonial angle:** Jensen's survey ranked mean gonial angle highest in Caucasians, then Chinese, Inuit and African groups, and lowest in Australian Aboriginal and American Indian samples. sciencedirect
- **Mandibular plane angle:** African-American norms in Philadelphia were SN–GoGn 32.0 ± 6.4° and FMA 25.2 ± 5.6°, close to the Steiner/Tweed values. springer
- **Eye spacing:** intercanthal distance is broadly similar across groups (≈32–34 mm), with Black adults slightly larger. sciencedirect Canthal tilt also differs. East Asian eyes often have a higher lateral canthus, and attractive East Asian female composites showed *less* extreme tilt than the population typical, the reverse of the Caucasian pattern. incels
- **Cheekbones:** ideal Chinese and Japanese female composites had flatter cheekbones and more obtuse mandibular angles (Rhee & Lee 2010). sciencedirect

Recommendations

- **Use these as targets (in order of reliability):**
  1. Nasofrontal 115–130°
  2. Cervicomental 105–120°
  3. G–Sn–Pg ≈168° ± 5°
  4. SN–GoGn 32° ± 4° / FMA 25° ± 5°
  5. Gonial angle within about ±7° of your sex- and ethnicity-specific mean
  6. A mildly positive canthal tilt (≈4–8°)
- **Use averageness ratios (36% / 46%) instead of phi** if you want a proportion rule.
- **Treat the non-academic metrics as descriptive, not diagnostic:** jaw frontal angle, cheekbone height %, ramus/mandible ratio, total FWHR and FaceIQ's eye separation ratio. Different apps define them differently, and their "ideal" bands are derived from curated celebrity or model faces.
- **For any surgical or orthodontic decision, use population-specific norms** taken from radiographs or calibrated photos, not phone selfies. Lens distortion and head tilt change angles by several degrees.

Caveats

- **"Ideal" is not "normal":** clinical norms (Steiner, Tweed, Jarabak, Farkas) describe average or normal-occlusion samples. Attractiveness values (Powell & Humphreys, Liu, Rhee) come from small, selected, mostly Caucasian samples. Neither is a medical threshold, and values outside these ranges are not pathological.
- **Small and secondary samples:** several sex-specific numbers come from single studies with small samples, e.g., 21 men and 21 women in Liu 2023, oup and 37 men and 51 women in Carré & McCormick. realsmile Some are cited here via secondary sources: Uzun 2014 and the Sommerville neck angles via Qoves, the Carré means via an FWHR calculator site, and Volpe & Ramirez via a blog. Verify these before relying on them.
- **Measurement conventions conflict:** different landmarks (glabella vs nasion, brow vs eyelid, interior angle vs supplement) can shift values by 5–20° or 0.1–0.2 ratio units. Always confirm the definition before comparing your number to a range.
- **Non-academic ranges:** the FaceIQ ranges reported here were read from its public celebrity pages. They vary slightly between pages and were not published with any methodology. The female facial-thirds bands even contradict Farkas's anthropometric norms, since they put the lower third below the middle third. faceiqlabs faceiqlabs